

Embedded projects 2

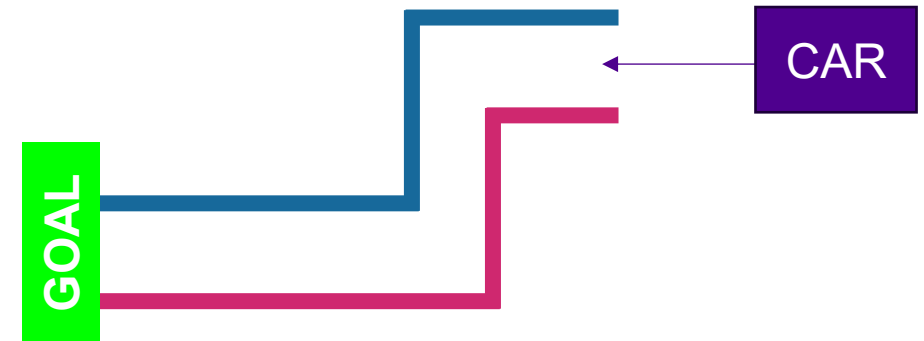
Weekly exercises

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Exercise 1 – Calibrating the sensor

- Attach the color sensor to your car
- Make sure you are able to calibrate your car to detect three different colors
- For calibration, use your joystick button to loop through the colors you want your car to detect (use RGB or CMY colors)
 - 1st press of button will start the calibration and detect the first color
 - 2nd press is the second etc...
 - After the third one the car will return to the “normal” ESP mode
- Remove the joystick driving functionality from your code, but keep the `attachInterrupt()` function to handle button presses.

Exercise 2 – Color maze



- Make a code, which drives your car through a simple maze
 - Your car should be able to drive through the maze, based on the color sensor information
 - Car is controlled from the website with the Drive commands
 - Change the drive forward 5 cm command to drive 50 cm instead **OR** Drive until the goal is reached
 - You must calibrate your colors at the start (as explained in exercise 1)
 - You must navigate primarily based on color sensor values, not encoder pulses
- At the corner of this slide, there is one example layout of the maze
 - There will be three different colors; your car should stay between the two lines and stop at the end
 - In your code, assume that the goal direction is either north, west... based on the maze
 - You can start testing closer to the goal and once you are satisfied move to the starting point
- The mazes will be printed on A3 paper, which you should be able to find somewhere in the lab